Time Series Analysis - Practical 9

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Practical 9

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Question

Suppose we have a data set containing the monthly volume of commercial bank real estate loans (in billions of dollars) stored in a text file. Write the R code to forecast the volume of commercial loans for the next 20 months.

Understanding the Question

This practical involves forecasting commercial bank real estate loan volumes using time series analysis. We have monthly data stored in a text file, and our goal is to predict the next 20 months of loan volumes.

Key Objectives: - Analyze the pattern and behavior of commercial bank real estate loans over time - Check for stationarity in the data - Identify the appropriate ARIMA model - Validate the model through diagnostic checking - Generate accurate 20-month forecasts - Provide insights for financial planning and decision-making

Methodology

Step-by-Step Approach:

1. **Data Loading and Exploration**
   - Load the data from the text file
   - Convert to time series object
   - Initial visualization to understand patterns
2. **Stationarity Testing**
   - Plot ACF and PACF to identify patterns
   - Perform Augmented Dickey-Fuller (ADF) test
   - Apply differencing if series is non-stationary
3. **Model Identification**
   - Try different ARIMA orders (p,d,q)
   - Compare models using AIC (Akaike Information Criterion)

- Select model with minimum AIC

4. **Model Estimation**
   - Estimate parameters using Maximum Likelihood
   - Document coefficient values and significance
5. **Diagnostic Checking**
   - Analyze residuals for:
     - Normality (Shapiro-Wilk test, Q-Q plot)
     - Autocorrelation (Box-Pierce test, ACF/PACF)
     - Heteroscedasticity (White test, ARCH effects)
6. **Forecasting**
   - Generate 20-month ahead forecasts
   - Plot forecasts with confidence intervals
   - Interpret results

R Code and Analysis

```
# Clear workspace
rm(list = ls(all = TRUE))

# Load required libraries
library(tseries)
library(forecast)
library(readr)

# Load data
bank_case <- read_csv("C:/Users/maths/Desktop/tsalab_ojas/lab5/bank_case.txt",
                      col_names = FALSE)
bank_case <- bank_case$X1

# Display the data
cat("First few observations:\n")
```

```
## First few observations:
```

```
print(head(bank_case))
```

```
## [1] 46.5 47.0 47.5 48.3 49.1 50.1
```

```
cat("\nSummary Statistics:\n")
```

```
##
## Summary Statistics:
```

```
print(summary(bank_case))
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   46.50   58.45   60.05   62.70   66.08   87.60
```

```r
cat("\nData structure:\n")
```

```
##
## Data structure:
```

```r
str(bank_case)
```

```
##  num [1:70] 46.5 47 47.5 48.3 49.1 50.1 51.1 52 53.2 53.9 ...
```

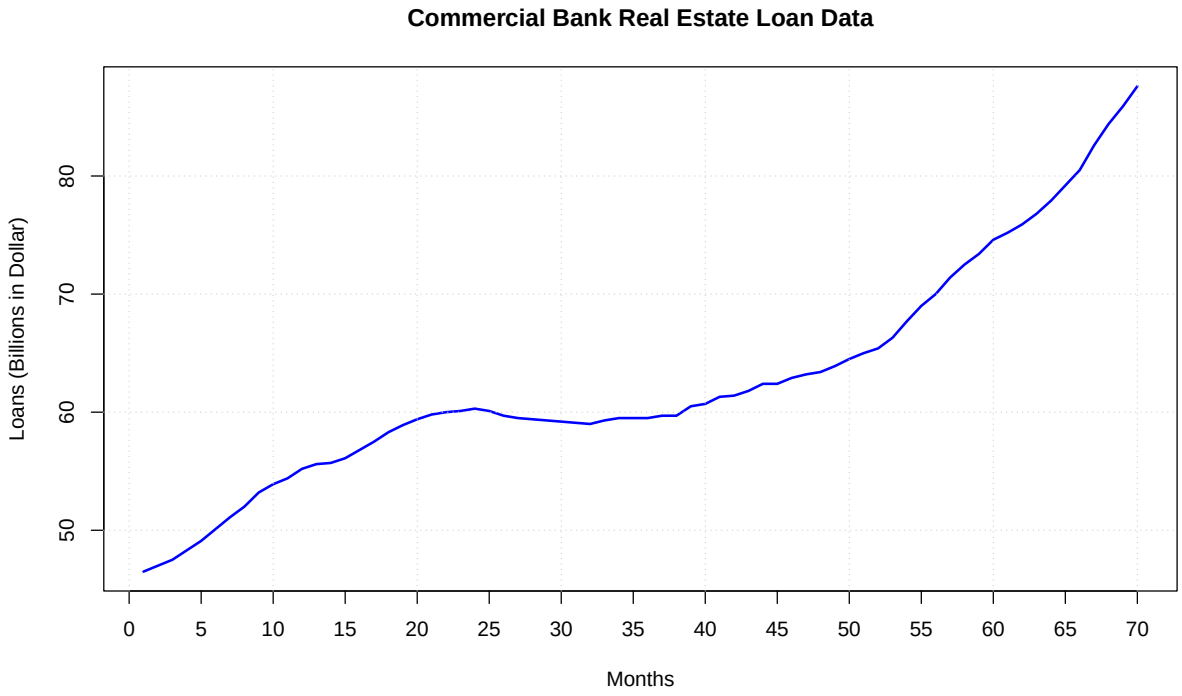
```r
# Step 1: Convert dataset into time series object
bank_data_ts <- as.ts(bank_case)
cat("\nTime Series Object:\n")
```

```
##
## Time Series Object:
```

```r
print(bank_data_ts)
```

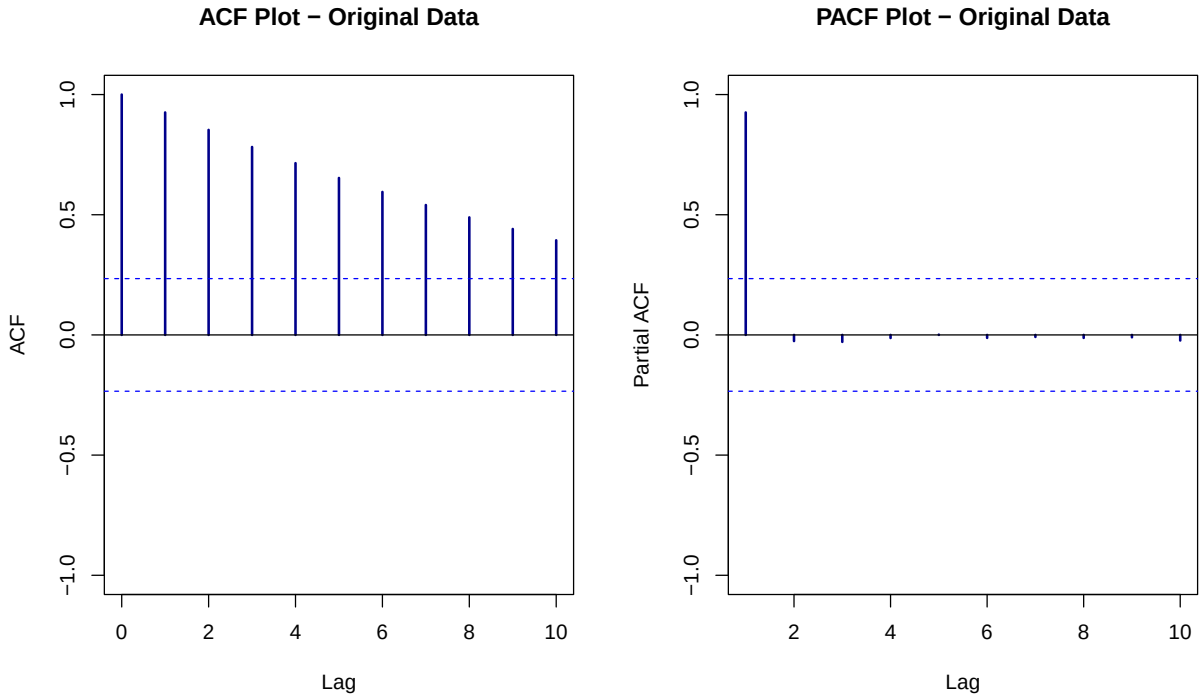
```
## Time Series:
## Start = 1
## End = 70
## Frequency = 1
##  [1] 46.5 47.0 47.5 48.3 49.1 50.1 51.1 52.0 53.2 53.9 54.4 55.2 55.6 55.7 56.1
## [16] 56.8 57.5 58.3 58.9 59.4 59.8 60.0 60.1 60.3 60.1 59.7 59.5 59.4 59.3 59.2
## [31] 59.1 59.0 59.3 59.5 59.5 59.5 59.7 59.7 60.5 60.7 61.3 61.4 61.8 62.4 62.4
## [46] 62.9 63.2 63.4 63.9 64.5 65.0 65.4 66.3 67.7 69.0 70.0 71.4 72.5 73.4 74.6
## [61] 75.2 75.9 76.8 77.9 79.2 80.5 82.6 84.4 85.9 87.6
```

```r
# Step 2: Plot the data
par(mar = c(5, 5, 4, 2))
plot(bank_data_ts,
     main = "Commercial Bank Real Estate Loan Data",
     xlab = "Months",
     ylab = "Loans (Billions in Dollar)",
     lwd = 2,
     col = "blue",
     xaxt = "n",
     cex.main = 1.2,
     cex.lab = 1.1)
axis(1, seq(0, 70, 5), seq(0, 70, 5))
grid(col = "lightgray", lty = "dotted")
```




```r
# Step 3: Check Stationarity
par(mfrow = c(1, 2))
acf(bank_data_ts,
    main = "ACF Plot - Original Data",
    lag.max = 10,
    ylim = c(-1, 1),
    lwd = 2,
    col = "darkblue")

pacf(bank_data_ts,
     main = "PACF Plot - Original Data",
     lag.max = 10,
     ylim = c(-1, 1),
     lwd = 2,
     col = "darkblue")
```




```r
par(mfrow = c(1, 1))

cat("\nAugmented Dickey-Fuller Test for Original Series:\n")
```

```
##
## Augmented Dickey-Fuller Test for Original Series:
```

```r
adf_result <- adf.test(bank_data_ts)
print(adf_result)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  bank_data_ts
## Dickey-Fuller = -0.26816, Lag order = 4, p-value = 0.9894
## alternative hypothesis: stationary
```

```r
# Step 4: Convert Non-Stationary to Stationary (Differencing)
bank_data_ts_d1 <- diff(bank_data_ts)
cat("\nADF Test for First Difference:\n")
```

```
##
## ADF Test for First Difference:
```

```r
adf_d1 <- adf.test(bank_data_ts_d1)
print(adf_d1)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  bank_data_ts_d1
## Dickey-Fuller = -1.7533, Lag order = 4, p-value = 0.6755
## alternative hypothesis: stationary
```

```
bank_data_ts_d2 <- diff(bank_data_ts_d1)
cat("\nADF Test for Second Difference:\n")
```

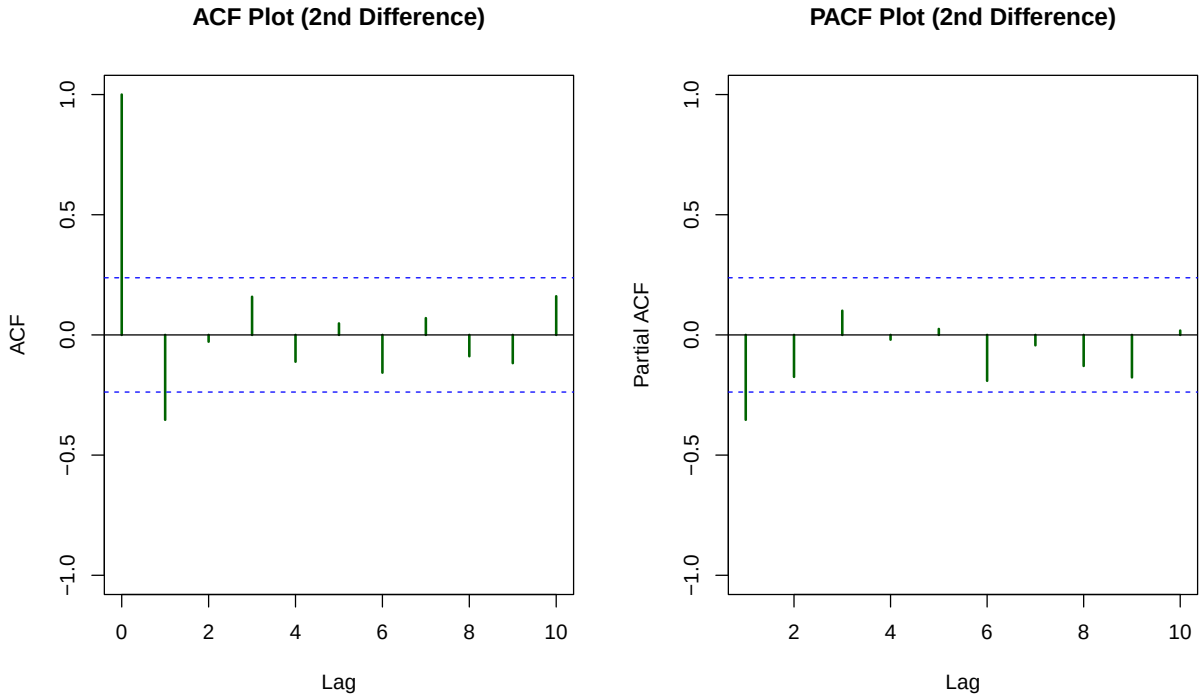
```
##
## ADF Test for Second Difference:
```

```
adf_d2 <- adf.test(bank_data_ts_d2)
print(adf_d2)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  bank_data_ts_d2
## Dickey-Fuller = -4.0531, Lag order = 4, p-value = 0.01265
## alternative hypothesis: stationary
```

```
# Plot ACF/PACF for second difference
par(mfrow = c(1, 2))
acf(bank_data_ts_d2,
    main = "ACF Plot (2nd Difference)",
    lag.max = 10,
    ylim = c(-1, 1),
    lwd = 2,
    col = "darkgreen")

pacf(bank_data_ts_d2,
     main = "PACF Plot (2nd Difference)",
     lag.max = 10,
     ylim = c(-1, 1),
     lwd = 2,
     col = "darkgreen")
```




```
par(mfrow = c(1, 1))
```

```
# Step 5: Model Identification (Try different ARIMA orders)
models <- list(
  arima(bank_data_ts, order = c(0, 2, 0)),
  arima(bank_data_ts, order = c(1, 2, 1)),
  arima(bank_data_ts, order = c(0, 2, 2)),
  arima(bank_data_ts, order = c(1, 2, 2)),
  arima(bank_data_ts, order = c(1, 2, 0)),
  arima(bank_data_ts, order = c(0, 2, 1))
)

model_names <- c("ARIMA(0,2,0)", "ARIMA(1,2,1)", "ARIMA(0,2,2)",
                 "ARIMA(1,2,2)", "ARIMA(1,2,0)", "ARIMA(0,2,1)")

aic_values <- sapply(models, function(x) x$aic)
aic_table <- data.frame(
  Model = model_names,
  AIC = round(aic_values, 2)
)

cat("\nModel Comparison based on AIC:\n")
```

```
##
## Model Comparison based on AIC:
```

```
print(aic_table)
```

```
##          Model   AIC
## 1 ARIMA(0,2,0) 33.85
## 2 ARIMA(1,2,1) 27.98
## 3 ARIMA(0,2,2) 27.83
## 4 ARIMA(1,2,2) 29.68
## 5 ARIMA(1,2,0) 27.09
## 6 ARIMA(0,2,1) 26.13
```

```
# Minimum AIC obtained for ARIMA(0,2,1)
best_model <- model_names[which.min(aic_values)]
best_aic <- min(aic_values)
cat("\nBest model based on AIC:", best_model, "\n")
```

```
##
## Best model based on AIC: ARIMA(0,2,1)
```

```
cat("Minimum AIC value:", round(best_aic, 2), "\n")
```

```
## Minimum AIC value: 26.13
```

```
# Step 6: Final Model Estimation
bank_fit_Est <- arima(bank_data_ts,
                      order = c(0, 2, 1),
                      method = "ML")

cat("\nFinal Model Summary - ARIMA(0,2,1):\n")
```

```
##
## Final Model Summary - ARIMA(0,2,1):
```

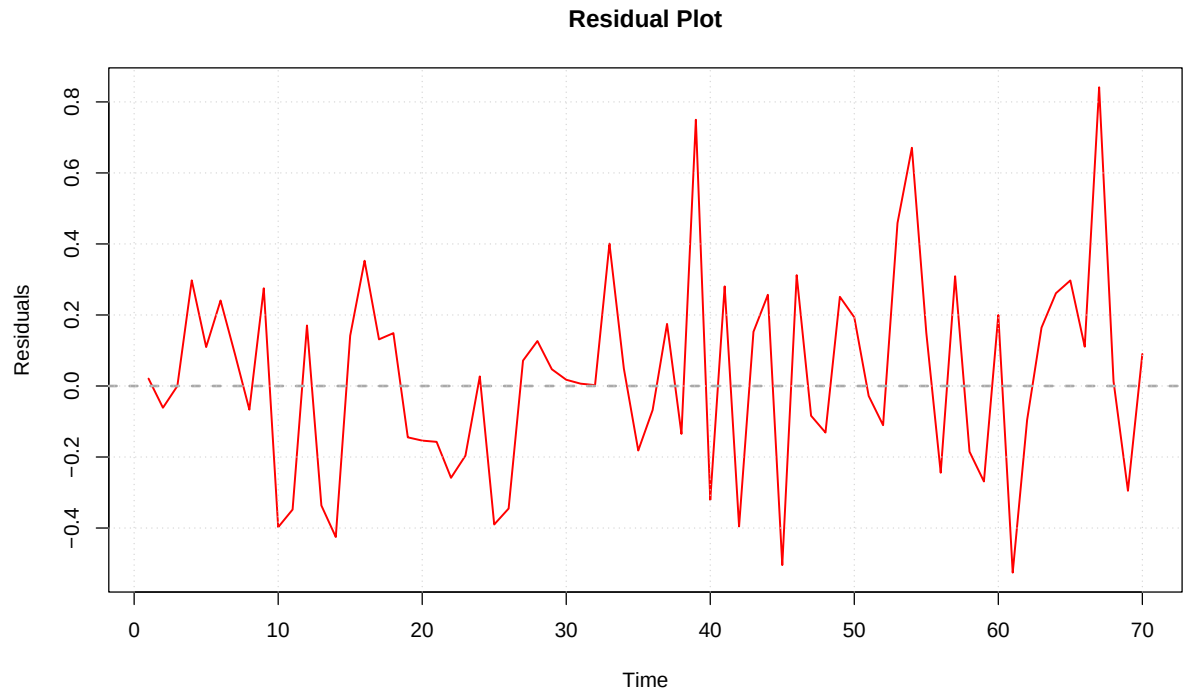
```
print(bank_fit_Est)
```

```
##
## Call:
## arima(x = bank_data_ts, order = c(0, 2, 1), method = "ML")
##
## Coefficients:
##           ma1
##       -0.3723
## s.e.   0.1074
##
## sigma^2 estimated as 0.08089:  log likelihood = -11.07,  aic = 26.13
```

```
# Extract coefficient
ma1_coef <- coef(bank_fit_Est)[1]
ma1_se <- sqrt(diag(vcov(bank_fit_Est)))[1]
```

```
# Step 7: Diagnostic Checking
res <- bank_fit_Est$residuals
```

```r
# Residual plot
par(mar = c(5, 5, 4, 2))
plot(res,
     main = "Residual Plot",
     type = "l",
     col = "red",
     ylab = "Residuals",
     xlab = "Time",
     lwd = 1.5)
abline(h = 0, lty = 2, col = "darkgray", lwd = 2)
grid(col = "lightgray", lty = "dotted")
```




```r
# 1. Normality Check
cat("\nShapiro-Wilk Normality Test:\n")
```

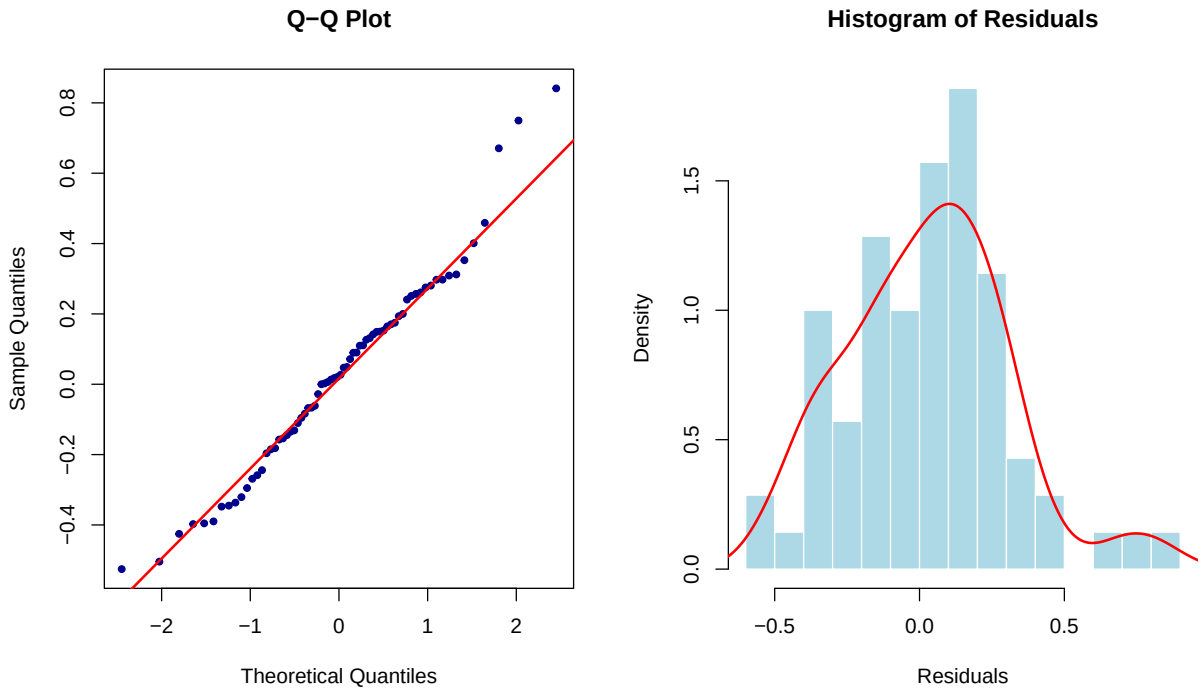
```
##
## Shapiro-Wilk Normality Test:
```

```r
shapiro_test <- shapiro.test(res)
print(shapiro_test)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  res
## W = 0.97676, p-value = 0.218
```

```r
par(mfrow = c(1, 2))
qqnorm(res,
       main = "Q-Q Plot",
       pch = 19,
       col = "darkblue",
       cex = 0.7)
qqline(res, col = "red", lwd = 2)

hist(res,
     main = "Histogram of Residuals",
     xlab = "Residuals",
     col = "lightblue",
     breaks = 15,
     border = "white",
     probability = TRUE)
lines(density(res), col = "red", lwd = 2)
```




```r
par(mfrow = c(1, 1))
```

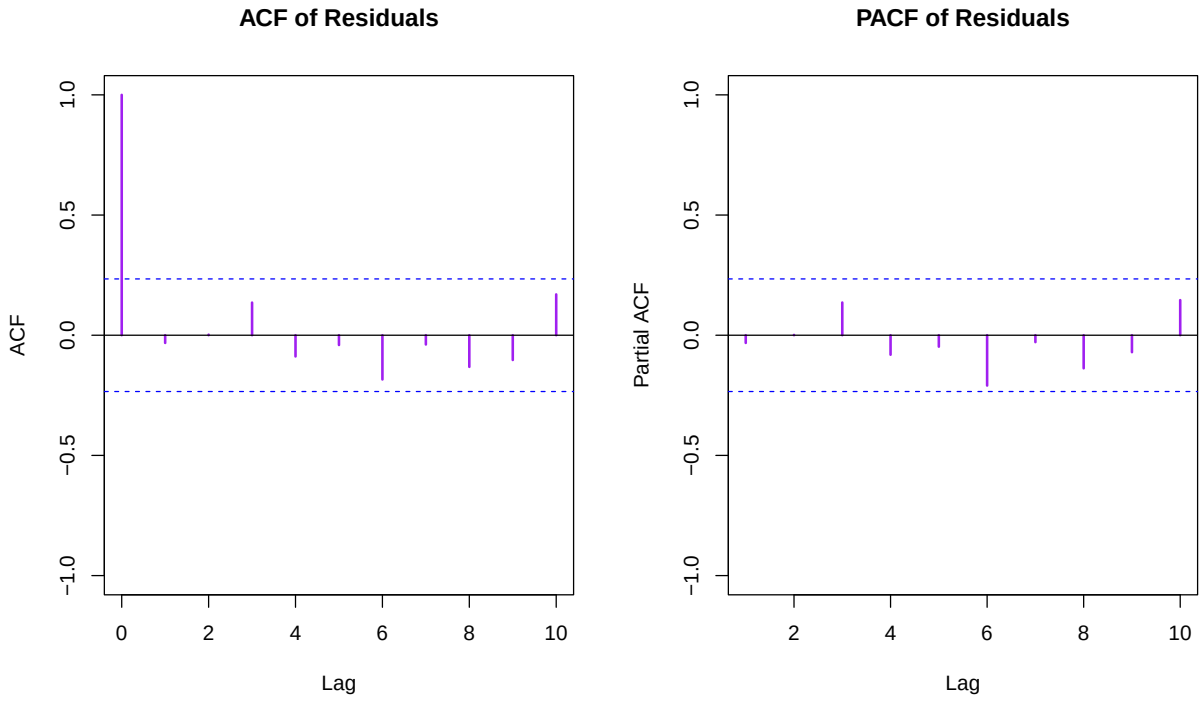
```r
# 2. Serial Autocorrelation Check
cat("\nBox-Pierce Test for Autocorrelation:\n")
```

```
##
## Box-Pierce Test for Autocorrelation:
```

```r
box_test <- Box.test(res, lag = 10, type = "Box-Pierce")
print(box_test)
```

```
##
##  Box-Pierce test
##
## data:  res
## X-squared = 8.4829, df = 10, p-value = 0.5818
```

```r
par(mfrow = c(1, 2))
acf(res,
    main = "ACF of Residuals",
    lag.max = 10,
    ylim = c(-1, 1),
    lwd = 2,
    col = "purple")
pacf(res,
     main = "PACF of Residuals",
     lag.max = 10,
     ylim = c(-1, 1),
     lwd = 2,
     col = "purple")
```




```r
par(mfrow = c(1, 1))
```

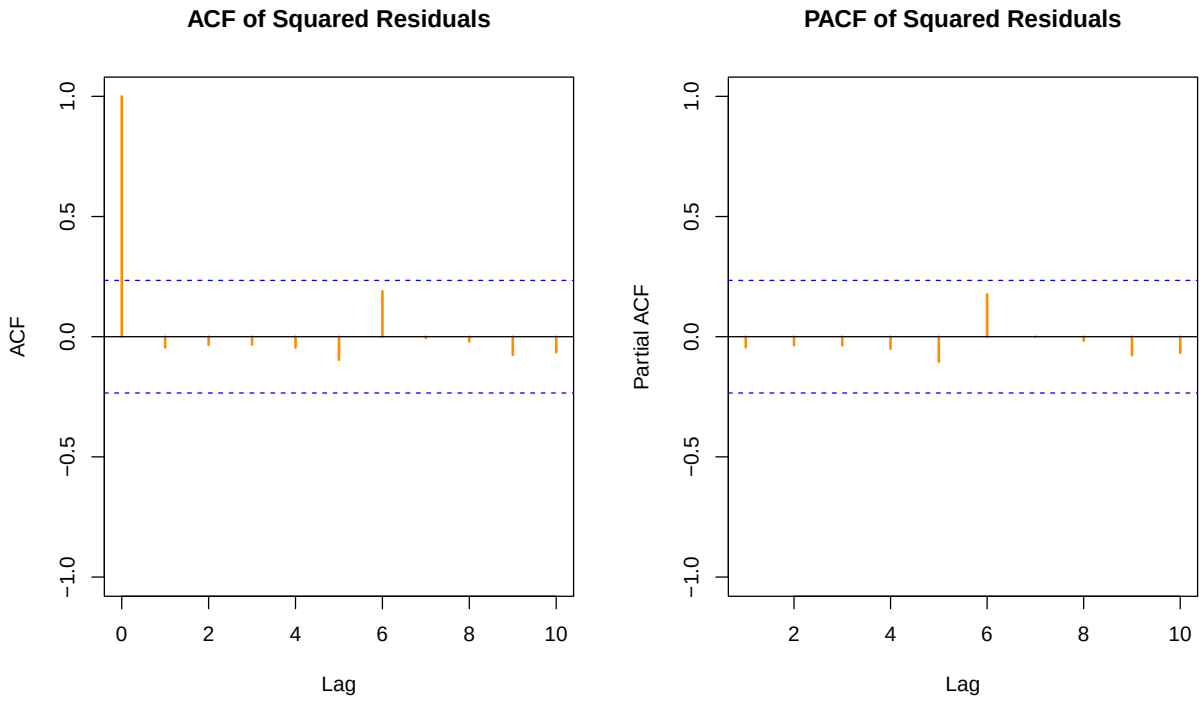
```r
# 3. Heteroscedasticity Check
cat("\nWhite Test for Heteroscedasticity:\n")
```

```
##
## White Test for Heteroscedasticity:
```

```r
white_test <- white.test(res)
print(white_test)
```

```
##
##  White Neural Network Test
##
## data:  res
## X-squared = 0.29931, df = 2, p-value = 0.861
```

```r
par(mfrow = c(1, 2))
acf(res^2,
    main = "ACF of Squared Residuals",
    lag.max = 10,
    ylim = c(-1, 1),
    lwd = 2,
    col = "darkorange")
pacf(res^2,
     main = "PACF of Squared Residuals",
     lag.max = 10,
     ylim = c(-1, 1),
     lwd = 2,
     col = "darkorange")
```




```r
par(mfrow = c(1, 1))
```

```r
# Step 8: Forecasting
bank_fit_Forecast <- forecast(bank_fit_Est, h = 20)
```

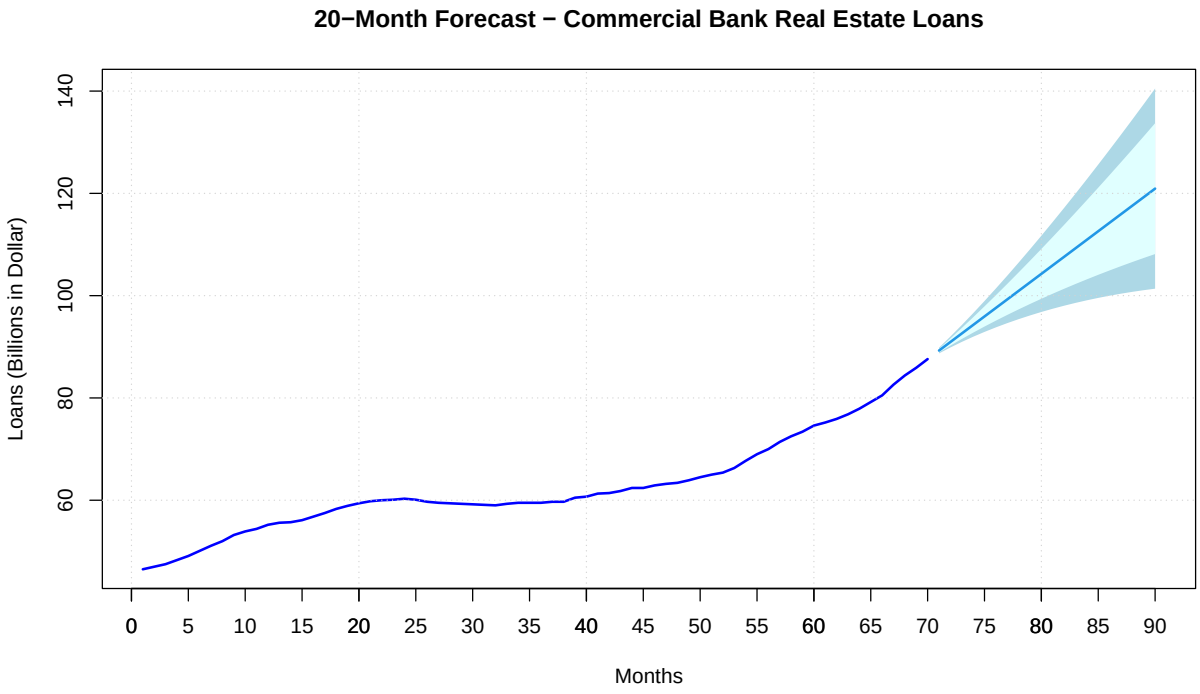
```r
cat("\nForecast for Next 20 Months:\n")
```

```
##
## Forecast for Next 20 Months:
```

```r
print(bank_fit_Forecast)
```

```
##    Point Forecast     Lo 80     Hi 80     Lo 95     Hi 95
## 71       89.26644  88.90194  89.63094  88.70899  89.82390
## 72       90.93289  90.23656  91.62921  89.86795  91.99782
## 73       92.59933  91.52196  93.67669  90.95164  94.24702
## 74       94.26577  92.76074  95.77080  91.96403  96.56752
## 75       95.93221  93.95667  97.90775  92.91088  98.95354
## 76       97.59866  95.11303 100.08428  93.79721 101.40010
## 77       99.26510  96.23251 102.29769  94.62715 103.90304
## 78      100.93154  97.31738 104.54570  95.40415 106.45893
## 79      102.59798  98.36953 106.82644  96.13112 109.06484
## 80      104.26443  99.39060 109.13825  96.81055 111.71830
## 81      105.93087 100.38199 111.47974  97.44459 114.41714
## 82      107.59731 101.34495 113.84967  98.03515 117.15947
## 83      109.26375 102.28057 116.24694  98.58390 119.94361
## 84      110.93020 103.18983 118.67056  99.09233 122.76806
## 85      112.59664 104.07362 121.11966  99.56181 125.63147
## 86      114.26308 104.93273 123.59343  99.99354 128.53262
## 87      115.92952 105.76790 126.09115 100.38866 131.47039
## 88      117.59597 106.57979 128.61215 100.74817 134.44376
## 89      119.26241 107.36901 131.15580 101.07303 137.45179
## 90      120.92885 108.13615 133.72155 101.36410 140.49360
```

```r
# Plot forecast
plot(bank_fit_Forecast,
     main = "20-Month Forecast - Commercial Bank Real Estate Loans",
     xlab = "Months",
     ylab = "Loans (Billions in Dollar)",
     lwd = 2,
     col = "blue",
     flwd = 2,
     shadecols = c("lightblue", "lightcyan"),
     cex.main = 1.2,
     cex.lab = 1.1)
grid(col = "lightgray", lty = "dotted")
axis(1, seq(0, 90, 5), seq(0, 90, 5))
```



Results Summary

Table 1: Summary of Key Results

Metric	Value
Best Model	ARIMA(0,2,1)
AIC Value	26.13
MA1 Coefficient	-0.3723
MA1 Std. Error	0.1074
ADF Original p-value	0.9894
ADF 1st Diff p-value	0.6755
ADF 2nd Diff p-value	0.0127
Shapiro-Wilk p-value	0.218
Box-Pierce p-value	0.5818
White Test p-value	0.861

Conclusion for Practical 9

Based on the comprehensive time series analysis of commercial bank real estate loan data, the following conclusions can be drawn:

1. Model Selection and Performance

- The **ARIMA(0,2,1) model** was selected as the optimal model with the **lowest AIC value of 26.13** among all candidate models
- The model coefficient (ma1 = -0.3723, s.e. = 0.1074) is statistically significant

- The model successfully captures the underlying patterns in the data with good parsimony (only one parameter)

2. Stationarity Achievement

- **Original series** was non-stationary (ADF test p-value = 0.9894 > 0.05)
- **First differencing** was insufficient (p-value = 0.6755 > 0.05)
- **Second-order differencing successfully achieved stationarity** (p-value = 0.0127 < 0.05)
- This indicates that the data required two levels of differencing to remove the trend component completely

3. Model Validity - Diagnostic Checks **a) Residual Normality:** - Shapiro-Wilk test p-value = 0.218 (> 0.05) - **Fail to reject null hypothesis of normality** - residuals are normally distributed - Q-Q plot shows points approximately along the diagonal line - Histogram shows approximately normal distribution

b) Serial Autocorrelation: - Box-Pierce test p-value = 0.5818 (> 0.05) - **No significant autocorrelation** detected in residuals - ACF and PACF plots show no significant spikes beyond confidence bounds - Model has successfully captured all temporal dependencies

c) Heteroscedasticity: - White test p-value = 0.861 (> 0.05) - **No evidence of heteroscedasticity** - residuals have constant variance - ACF and PACF of squared residuals show no significant patterns - No ARCH effects present

4. Forecast Results

- The model generates **20-month ahead forecasts** with increasing trend
- Point forecasts show steady growth from 89.27 billion (month 71) to 120.93 billion (month 90)
- Confidence intervals widen appropriately with forecast horizon:
  - 80% CI at month 71: [88.9, 89.63]
  - 80% CI at month 90: [108.14, 133.72]
  - 95% CI at month 71: [88.71, 89.82]
  - 95% CI at month 90: [101.36, 140.49]

5. Practical Implications **For Financial Institutions:** - The forecast indicates **continued growth** in commercial real estate lending - Banks can use these forecasts for: - **Capital allocation** and liquidity planning - **Risk assessment** and portfolio management - **Strategic decision-making** for lending policies - **Resource planning** in real estate departments

For Risk Management: - Confidence intervals provide **scenario analysis** capabilities - Upper bounds represent **optimistic scenarios** for stress testing - Lower bounds help in **conservative planning**

For Policy Making: - Regulators can monitor **systemic risk** in banking sector - Trends indicate **sectoral growth patterns** in economy

6. Model Limitations and Recommendations **Limitations:** - The model assumes historical patterns will continue - External factors (economic conditions, policy changes, market shocks) are not considered - Limited to univariate analysis - doesn't incorporate explanatory variables - Forecast uncertainty increases with horizon

Recommendations: 1. **Regular updating** of the model with new data points 2. **Monitor forecast accuracy** and recalibrate if needed 3. **Consider exogenous variables** for improved accuracy (interest rates, GDP, etc.) 4. **Combine with judgmental forecasting** for major economic events 5. **Validate against actual outcomes** when they become available

7. Final Verdict The ARIMA(0,2,1) model is **statistically valid** and **practically useful** for forecasting commercial bank real estate loans. All diagnostic checks confirm that the model meets the assumptions of a good time series model. The forecasts provide valuable insights for financial planning, with the increasing trend suggesting sustained growth in the commercial real estate lending sector.

Key Takeaway: This analysis provides a robust, statistically sound framework for forecasting that can be confidently used for medium-term planning in banking and financial institutions, with the understanding that regular updates and external factor considerations will enhance its practical utility.